MILITARY SPECIFICATION

INDICATOR, THRUST, ENGINE TESTING, TEST STAND,
AIRCRAFT ENGINE, A/M37 SERIES

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of thrust indicator used in the A/M37 series jet engine test stand.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-V-173	Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronics, and Associated Equipment)
MIL-E-4158	Electronic Equipment Ground, General Requirement for
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for

STANDARDS

Military

MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-808	Finishes, Protective, And Codes, For Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Report, Preparation of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWING

Air Force

66C64622 Housing, Instrument-Control Cab

3. REQUIREMENTS

3.1 Qualification. Thrust indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4)

3.2 Description. The thrust indicator covered by this specification shall be used in the Air Force A/M37 series jet engine test stands for measuring engine thrust up to 35,000 pounds.

3.3 Design. The requirements of MIL-E-4158 apply except as specifically changed herein.

a. Paragraph 3.2.30.1.1 operating ambient of instrumentation is 20° to 130° Fahrenheit (F).

b. Paragraph 3.2.30.2.a inclined to any angle up to 20 degrees from normal position.

c. Paragraph 3.2.30.2.b vibration shall be that normally encountered in equipment towed over unimproved roads mounted in trailers that are not equipped with springs or shock absorbers. This trailer is defined as Type II, Group C mobility per MIL-M-6090.

d. Paragraph 3.2.30.3 electrical power shall be that normally supplied by a standard Air Force MD-3 generator set. Voltage and frequency shall be that shown on the electrical Schematic.

3.3.1 Design features. The thrust indicator shall be all solid state design with modular plug in circuit boards except for the display which shall be plug in type LED's. The display shall be continuous and shall not update unless the measurement is different from that counted during measuring cycle. The instrument shall be capable of displaying a selected input and continue to convert analog inputs to subsequent decimal display concurrently. The indicator shall be compatible with the load cells, wiring, physical dimensions and operating characteristics of the A/M37 series engine test stand.

3.3.1.1 Range and accuracy. The indicator shall have a range of 0 to 35,000 pounds with an accuracy of plus or minus 0.1 percent of reading or plus or minus 10 pounds, whichever is greater, throughout the operating temperature range of 20° to 130°F. This accuracy shall include long-term stability (30 days), drift, temperature effects and last digit ambiguity. If required to meet the accuracy, the indicator may be a dual range instrument with a range of 0 to 6,000 pounds and 6,000 pounds to 35,000 pounds.

3.3.1.2 Input. Input shall be obtained from the test cell wiring and two Revere Corporation of America 25,000 pound compression load cells Part Number 14C-41630-15, or equivalent, connected to sum the output of both cells. Input impedence shall be 20 megohms minimum.

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3.3.1.3 Power supply. The indicator shall contain the power supply for load cell excitation. The effects of power drift shall be minimized with self-compensating circuits or other means to meet the accuracies specified herein.

3.3.1.4 Zero balance adjustment shall be provided to zero the load cells output under no load. This adjustment shall be made with variable components in lieu of replacing fixed value components and shall be located on the face of the instrument.

3.3.1.5 Calibration adjustment shall be provided for full span calibration adjustment. This adjustment shall be made with variable components in lieu of replacing fixed value components and shall be located on the face of the instrument.

3.3.1.6 Range selector switch shall be provided if required to select the input for either the high range (6,000 to 35,000) or low range (0-6,000).

3.3.1.7 On-off power switch shall be provided on the face of the instrument.

3.3.1.8 Internal check circuits shall be provided to permit the operator to quickly determine the serviceability of the instrument.

3.3.1.8.1 Check/operate switch shall be provided for selecting either the check circuit mode or the operate mode located on the face of the instrument.

3.3.1.9 Physical mounting. The thrust indicator shall mount in the instrument panel of the A/M37B-6 series engine test stand console. The physical dimensions for the instrument panel housing are shown in Air Force Drawing 66064622-1. The face panel size shall be 6.00 to 6.10 inches wide by 6.40 to 6.5 inches high.

3.3.1.10 Electrical compatibility. The thrust indicator shall be electrically compatible with the electrical schematic shown in Figure 1.

3.3.1.11 Instrument case. The thrust indicator shall be provided with an instrument car or container to protect the instrument during shipping and use. The container may be either a permanent part of the instrument or may be removable.

3.3.1.12 Common mode rejection. The indicator shall be capable of rejecting noise (spurious and/or continuous) in the range of DC to 500 cycles from a level of 120 db down.

3.3.1.13 Sampling rate. The indicator shall have sampling rate from 1.2 to 3.0 seconds.

3.3.1.14 Warm up. The accuracy specified in paragraph 3.3.1.1 shall be obtainable after an initial 15-minute warm up maximum.

3.3.1.15 Reliability. The indicator shall have a mean-time-between-failures (MTBF) of 130 hours.

3.3.1.16 Maintainability. Any maintenance action on this indicator due to failure shall not exceed 3 hours including diagnosis, repair, replacement and calibration.

3.4 Finish. The indicator shall be finished per MIL-STD-808, color code USAF 514 (Instrument Black).

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4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections or tests where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing requirements specified herein are classified as follows

a. Qualification testing (see 4.4)

b. Acceptance testing (see 4.5)

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be conducted at standard atmospheric pressure within the temperature range of 70 to 80°F and a relative humidity of 60 to 70 percent. When tests are conducted with atmospheric pressure, temperature or relative humidity differing materially from above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Preliminary run-in and calibration. Preliminary operation of the indicator as may be recommended by the supplier shall be performed by the supplier prior to submission for tests. A certified log of all preliminary operating time, calibration data, et cetera, shall be prepared and included in the test report.

4.4 Qualification testing.

4.4.1 Qualification test sample. One indicator of each type to be supplied shall be subjected to the test specified in 4.4.3.

4.4.2 Qualification test report. After the qualification tests are complete, a test report in accordance with MIL-STD-831 shall be prepared.

4.4.2.1 Reliability and maintainability information. The following information shall be incorporated as an attachment to the qualification test report and shall be included as part of that report.

a. All failures, maintenance and other events recorded shall be identified by accumulated operating time, cycles or position in the test procedure as appropriate. Test conditions during the failures or irregular operations shall be recorded. There must be rationale to assure the MTBF requirement is met.

b. Description of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Description of the engineering reasoning verifying corrections made, to be made on production items, or proposed to be made that substantiate the predicted effectiveness of those corrections.

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d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. Verify maximum maintenance downtime by simulating 'worse case' failure, timing a technician in a fault discovered to operational ready simulation. Worse case failure shall be determined by the contractor unless directed otherwise by the Government inspector." A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to permit comparison to the personnel anticipated in similar field work.

4.4.3 Qualification tests. The qualification tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the individual tests (see 4.5.1).

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6.

a. Examination of Product (see 4.6.1)

b. Calibration Tests (see 4.6.2)

4.6 Test methods.

4.6.1 Examination of product. Each indicator shall be inspected to determine compliance with the requirements specified herein. Inspection shall also include workmanship, finish, markings, wire coding, and neatness of wiring.

4.6.2 Calibration tests. Each indicator shall be checked for calibration to determine compliance with the accuracy requirements specified herein. Calibration points shall be at 10, 20, 40, 50, 60, 75, 90 and 100 percent of each range.

4.6.3 Compatibility test. The thrust indicator shall be installed in an A/M37T-00 or A/M37T-20 series engine test stand by the procuring activity to verify the compatibility of the thrust indicator with the test stand. The thrust indicator shall be calibrated with an A/M37 portable thrust calibration kit. The thrust indicator shall remain installed in the test stand for a minimum of 2 hours of actual engine testing. At the end of this period, the calibration shall be repeated. This calibration shall be made with all test stand electrical systems operating. Failure to meet the accuracy requirement of paragraph 3.3.1 shall be cause for rejection.

4.6.4 Self-check-test. The self-check test circuit shall verify that the requirement of paragraph 3.3.1.8 has been met.

4.6.5 Long-term-stability. The indicator shall not require calibration during first article test until after the long-term stability has been checked. The indicator shall retain its calibration for 300 hours of operation. The time shall include the operating time during environmental testing. Assume 300 hours of operation is equivalent to 30 days operation.

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4.6.6 Drift. The input voltage shall be varied from 105 volts to 125 volts. Calibration shall be checked with 105 volts and again at 125 volts while installed in the A/M37T-6 series engine test stand using the thrust calibrator. Failure to meet the accuracy requirements of paragraph 3.3.1 shall be cause for rejection.

4.6.7 Environmental tests. The thrust indicator shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-STD-810. Upon completion of each test, the indicator shall be subjected to the calibration test specified in 4.6.2. Failure of the indicator to operate and/or meet the accuracy specified during or after each exposure, failure of any item or component due to deterioration or formation of any undesirable condition that cannot be remedied by routine maintenance shall be cause for rejection.

4.6.7.1 Low temperature test. The indicator shall be subjected to low temperature storage in accordance with Method 502, Procedure I. The low temperature storage temperature shall be -65°F for 8 hours. The temperature shall be raised to 32° ±3°F and stabilized. The calibration test specified in 4.6.2 shall then be conducted.

4.6.7.2 High temperature test. The indicator shall be subjected to the high temperature storage in accordance with Method 501 Procedure I, except Step 3 shall be 8 hours. The temperature shall then be lowered to 130°F and stabilized. The calibration tests specified in 4.6.2 shall then be conducted.

4.6.7.3 Humidity test. The indicator shall be subjected to humidity test in accordance with Method 507, Procedure I, except Step 5 shall be 5 cycles (not less than 120 hours).

4.6.7.4 Fungus-resistance test. The indicator shall be subjected to fungus test in accordance with Method 508, Procedure I, or certification shall be given that the indicator neither supports nor is damaged by fungus and that all components have been treated with MIL-V-173 Moisture and Fungus-Resistant Varnish.

4.6.7.5 Vibration. The indicator shall be subjected to the vibration test in accordance with Method 514.1, Procedure X, and Table 514.1-VII. Assume 1000 miles for land travel mileage.

4.6.7.6 Reliability. A specified MTBF (C as defined in MIL-STD-781) of 130 hours shall be proven in accordance with Test Plan II of MIL-STD-781. All preproduction testing shall count as part of this test both timewise and failwise. Additional testing to satisfy Test Plan II requirements will be performed at Test Level A-1 (room temperature).

5. Preparation for delivery shall be as specified in the invitation for bids or request for quote.

6. NOTES

6.1 Intended use. The thrust indicator is intended for use in the A/M37 series jet engine test stands.

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6.2 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is AFLC CASO/LODS, Federal Center, Battle Creek MI 49016, and information pertaining to qualification of products may be obtained from that activity.

6.3 Supersession data. This specification includes the requirements of Purchase Description PD-EAAHA/MMH-71-248 dated 12 Apr 71.

6.4 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

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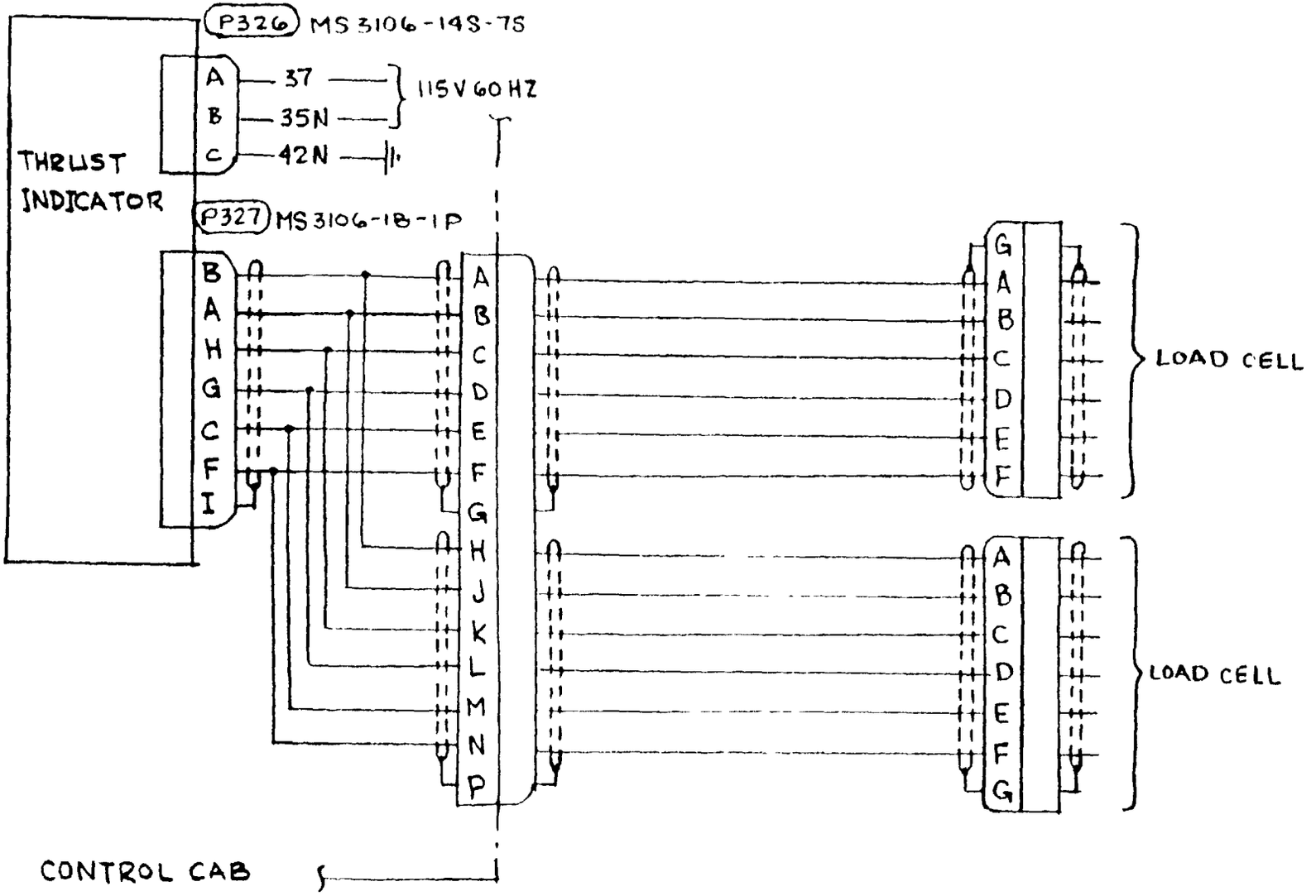


FIGURE 1 SYSTEM SCHEMATIC

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